A Primer on Labeled Property Graphs

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Abstract

Psychological data are inherently relational: they capture individuals embedded in dyads, groups, and institutions; behaviors unfolding across time; and constructs operating simultaneously at multiple levels of analysis. Yet the rectangular data formats that dominate psychological research impose tabular constraints that fragment, duplicate, or discard the relational structure that makes such data meaningful. This tutorial introduces labeled property graphs (LPGs) as a flexible and scalable database paradigm for modeling complex psychological data, and provides a step-by-step implementation guide using Neo4j and R. We illustrate the approach with the Speed-Dating replication dataset (Selterman et al., 2023), a publicly available resource whose dyadic, longitudinal, and multimodal structure makes it an ideal vehicle for demonstrating how LPGs preserve relational topology, handle structural missingness without numerical encoding, and support queries that rectangular formats cannot express without costly joins or data duplication. Compared to relational databases, LPGs offer three methodological advantages particularly relevant to psychological research: schema flexibility that accommodates evolving theoretical models, query performance that scales with local graph structure rather than total dataset size, and an ontological transparency that makes the researcher's assumptions about entities and relationships explicit and testable. All materials — including annotated code, data, and a reproducible workflow — are available via the Open Science Framework to facilitate adoption across subfields of psychology.

Keywords: Graph database, Network modeling, complex behavior

1 Introduction

Psychological data are relational by nature. For example, Guimerà, Uzzi, Spiro, and Amaral (2005) leveraged this relational nature of psychological data to understand the mechanisms of team assembly and how these mechanisms determine collaboration structure and team performance. Other instances include the analyses of the causal interplay between psychopathological symptoms (Borsboom & Cramer, 2013), the scientific cooperation to understand cyberbehavior (Serafin, Garcia-Vargas, García-Chivita, Caicedo, & Correra, 2019), the psychometric estimation of the correct number of dimensions in psychological and educational instruments (Golino & Epskamp, 2017), or the measurement of organizational climate (Menezes, Menezes, Moraes, & Pires, 2021). What these examples share, however, is that relational structure was an analytical construction rather than a database design — and it is precisely that distinction that this article addresses.

Relational databases have dominated psychological research for good reasons. They store data in tables where columns represent variables, rows represent cases, and relationships between entities are managed through the so-called "SQL" or Structured Query Language (Soto, 2025). Relational databases represent a paradigm that works well when data are rectangular and relationships are simple. However, as psychological phenomena grow more complex (i.e., spanning multiple levels of analysis, unfolding over time, and involving entities that relate to each other in qualitatively distinct ways) the tabular constraints of relational databases become a methodological liability rather than an asset. An alternative paradigm, broadly called "Not Only SQL" (NoSQL), includes systems designed for non-tabular, highly connected data. Among these, graph databases are particularly well-suited for psychological research because they treat relationships as first-class elements rather than implicit byproducts of table joins (Robinson, Webber, & Eifrem, 2015). In this article, we examine graph databases as a methodological paradigm where nodes and edges (rather than tables and joins) become the basic representational units for any behavior. One of the benefits that results from this database paradigm is that the behavior under scrutiny can be modeled as a complex system in a way that deviates from previous approaches (Guastello, Koopmans, & Pincus, 2009). Understanding why this database paradigm remains largely absent from psychological research requires first acknowledging what already exists.

Computational libraries such as `sna`, `igraph`, `qgraph`, or `psychonetrics` have made network analysis accessible to researchers working entirely within `R` (Luke, 2015), and their adoption in psychology is well-known (Mair, 2018). However, these libraries mainly address the statistical analysis of relational data, leaving the management of it under the hood. (i.e., the data must first be extracted from a rectangular format, reshaped into an edge list or adjacency matrix, and then passed to the analytic tool). Thus, such a workflow reconstructs relational structure at the analysis stage rather than preserving it from the moment of data collection. Graph database management systems, by contrast, store, query, and retrieve data in graph form natively, so that relational structure is not an analytical construction but a representational commitment. To our knowledge, this distinction has received no systematic treatment as a psychological method.

This article provides a step-by-step tutorial for implementing labeled property graphs (LPGs) in psychological research using `Neo4j` and `R`. We illustrate the approach with the Speed-Dating replication dataset (Selterman, Chagnon, & Mackinnon, 2023), a publicly available resource whose dyadic, longitudinal, and multimodal structure makes it an ideal vehicle for demonstrating how LPGs preserve relational topology, handle structural missingness without numerical encoding, and support queries that rectangular formats cannot express without costly joins or data duplication. The tutorial covers the full workflow: designing a graph schema from psychological theory, importing data into Neo4j, querying the graph with Cypher, connecting Neo4j to R via the neo2R package, and conducting statistical analysis and visualization using tidygraph and ggraph. All materials — including annotated Cypher and R code, data, and a reproducible RMarkdown file — are available in a public repository.

The remainder of this article is organized as follows. Section 2 introduces the conceptual foundations of labeled property graphs, contrasting their representational logic with that of relational databases and defining the core elements — nodes, edges, labels, and properties — that make them particularly well-suited for psychological data. Section 3 motivates the tutorial through the Speed-Dating replication dataset (Selterman et al., 2023), showing concretely how its dyadic, longitudinal, and multimodal structure exposes the limitations of rectangular formats. Section 4 presents the step-by-step tutorial itself, covering the full workflow from graph schema design and data import in Neo4j through querying with Cypher and statistical analysis and visualization in R via the neo2R, tidygraph, and ggraph packages. Section 5 discusses the broader methodological implications of LPGs for psychological research, with particular attention to structural missingness, scalability, and ontological transparency. Section 6 concludes with a discussion of limitations and directions for future work. All code, data, and a fully reproducible RMarkdown file are available via the Open Science Framework at [OSF link].

2 Graph-based databases: A gentle introduction

Any graph-based database relies on network basic terminology. Thus, a brief recap is useful here. Estrada (2011) defines a network as a collection of points (called nodes) joined together in pairs by lines (called arcs or edges) like those depicted in Figure 1. Despite this simplistic definition, networks provide a powerful framework to model any type of system from planets in a galaxy to neurons in the nervous system (Vazza & Feletti, 2020).

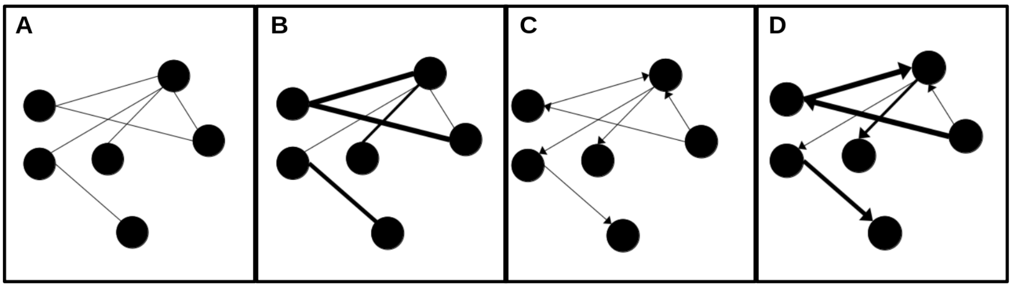

Fig. 1 A visual representation of four types of networks: A) non-directed unweighted network, B) non-directed weighted network, C) directed unweighted network, and D) directed weighted network.

In behavioral sciences, networks have been used to understand the mechanisms of team assembly and how these mechanisms determine collaboration structure and team performance (Guimerà et al., 2005). Graphs offer fundamental concepts for understanding how entities (nodes) and their relationships (edges) form interconnected data structures. From a data management viewpoint, the analysis of these data structures requires tools that go beyond the rigid tabular constraints of relational databases. As the concepts of graphs are thoroughly covered in introductory texts (Newman, 2010), these will not be revisited here. Instead, this article aims to illustrate how graph databases can enrich the methodological toolbox of psychological researchers and behavioral scientists, enabling analyses that embrace complexity, dynamic relationships, and multi-level contingencies (Robinson et al., 2015).

Robinson et al. (2015) define a graph database as a system that implements **C**reate, **R**ead, **U**pdate, and **D**elete (CRUD) operations on a graph data model, where entities are represented as nodes and relationships as edges like the one depicted in Figure 2 derived from the Speed-Dating dataset (Selterman et al., 2023) which was a replication study from the original article by Eastwick and Finkel (2008).

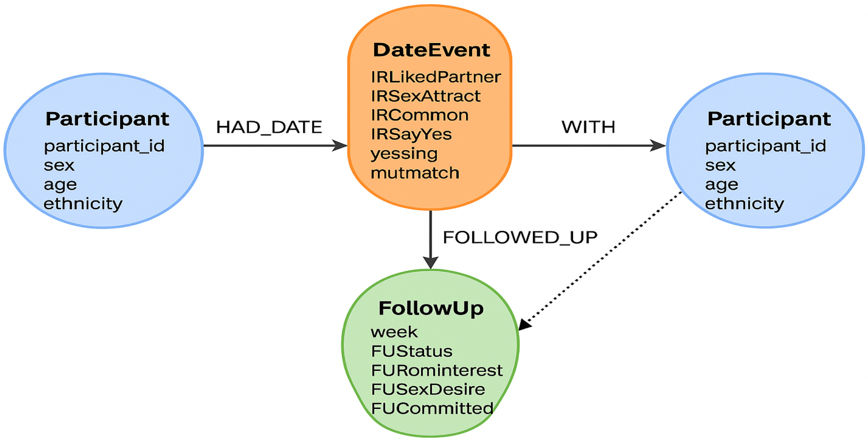

Fig. 2 A visual representation of a graph database that combines persons, companies, and routes

In contrast to relational databases (where relationships exist only implicitly through table joins) graph databases treat relationships as fundamental elements. In the context of the study conducted by Eastwick and Finkel (2008), the graph-based design enables researchers to traverse, query, and model highly interconnected data structures with exceptional efficiency and originality.

Figure 2 shows the data at three distinct levels: (1) stable participant characteristics (e.g., age, ethnicity, sex), (2) interaction-level impressions recorded during each brief date (e.g., attraction, perceived similarity, desire to say "yes"), and (3) longitudinal follow-up responses across subsequent weeks for pairs who continued interacting. These layers of data are represented as interconnected components of a single, coherent graph structure.

Voy a introducir un cambio en esta parte.

2.1 Entities as Nodes

`Participant` nodes represent individuals in the study and store stable demographic properties such as age, ethnicity, and sex (from the codebook variables DemAge, DemEthn, and sex_M1_F2). `DateEvent` nodes represent each dyadic interaction and store the full set of impression-formation ratings provided by participants during speed-dating (e.g., IRLikedPartner, IRSexAttract, IRCommon, IRSayYes, yessing, mutmatch). These correspond directly to the interaction-record variables in the dataset. `FollowUp` nodes represent week-level longitudinal responses, including relationship status, romantic interest, sexual desire, and commitment indicators (e.g., FUStatus, FURomInterest, FUSexDesire, FUCommitted), captured only for participants who matched and continued interacting.

2.2 Relationships as Edges

Edges define how these nodes are linked and encode psychological structure directly. `HAD_DATE` edges connect Participants to the DateEvents in which they took part, indicating the rater's perspective on that interaction. `WITH` edges connect the DateEvent back to the other participant, allowing the system to explicitly represent dyadic structure. `FOLLOWED_UP` edges link `Participants` to their `FollowUp` nodes, representing relational trajectories across weeks. Optionally, a `MUTUAL_MATCH_WITH` edge can be created between two `Participants` when the `mutmatch` property equals 1, providing a direct representation of reciprocal liking (i.e., a relationship that is invisible in rectangular data but naturally expressed in graph form).

2.3 Why This Matters for Psychological Research

This structure offers several methodological advantages. Psychological data retain their natural relational structure. Dyadic interactions are modeled as shared events rather than duplicated rows, preserving the topology of social and romantic processes as they occur in the real world. Temporal processes become navigable. `Follow-up` nodes can be traversed sequentially, enabling queries such as "How early do committed feelings emerge among mutually matched pairs?'' or "Does initial sexual attraction predict later romantic interest?''

In addition, missingness is handled structurally, not numerically. Participants who did not match simply lack `follow-up` nodes; no NA encoding is required. The model is inherently extensible. Additional constructs, such as personality traits or behavioral logs, can be attached as new nodes or properties without reconfiguring the existing database schema.

In sum, LPGs allow behavioral scientists to represent complex, multi-level psychological phenomena in a way that relational tables cannot. The Speed-Dating schema makes clear how impression formation, dyadic reciprocity, and longitudinal relationship development can be captured and analyzed within a single unified model.

3 Why graph databases as psychological method

Any graph database model leverages the so-called "labeled property graph." As a computational tool it paves the way for incorporating complex systems thinking in any discipline (Krakauer, 2019), and as a psychological method it opens the door for interdisciplinary collaboration. In psychology and behavioral sciences, complex systems thinking is not novel at all (Guastello et al., 2009; Luce, 1999). However, labeled property graph models and their elements have been largely overlooked. This gap coincides with recent reviews that promote other methodologies as potential solutions to increase intra and interdisciplinary interactions across subfields of behavior analysis in both basic and applied settings (Elcoro, Diller, & Correa, 2023).

As a psychological method, labeled property graph models share some similarities with the "nomological nets" coined by Cronbach and Meehl (1955). However, labeled property graph models (also known as knowledge graphs) represent a more general purpose toolkit that transcends the seminal conception of nomological nets (see Table 1).

Table 1: Knowledge graphs and nomological nets

Feature	Knowledge graphs	Nomological nets
Primary field	Computer science / AI / Data engineering	Psychology / Philosophy of Science
Primary goal	Data integration, search, automated reasoning.	Establishing construct validity (i.e., proving a theory).
Nature of nodes	Concrete entities (e.g., "Lee J. Cronbach", "University of Illinois").	Theoretical constructs (e.g., "introversion", "intelligence").
Nature of edges	Predicate or properties (e.g., "works_at", "collaborate_with").	Theoretical laws or empirical correlations.
Verification	Based on data consistency and retrieval accuracy	Based on statistical evidence (correlations and p-values)
Scale	Can contain billions of facts (e.g., Google knowledge graph)	Usually small, focused on a specific theoretical model

A methodolgical implication embedded in graph databases is their performance when handling network data structures (Robinson et al., 2015). In relational databases, join-intensive queries slow down as datasets grow, whereas graph databases maintain relatively stable performance —even with millions of nodes and edges. From a traditional perspective, the idea of working with huge datasets might seem implausible for most psychological researchers. Nonetheless, big data research is no longer outside professional considerations and that explains why big data has been recently reviewed as a psychological method (Vezzoli & Zogmaister, 2023). For example, previous works in network science has shown how to address complex behavioral phenomena such as the analysis of urban traffic at an individual level by tracking the movements of hundreds of thousands of drivers every two hours (Gonzalez, Hidalgo, & Barabasi, 2008). In cases like this, graph databases excel because queries operate on localized portions of the graph rather than scanning the entire dataset. If edges store information such as commuting distances, queries can be designed to retrieve specific conditions (e.g., individuals who commute less than 10 miles versus persons who commute more than 20 miles). Consequently, execution time depends only on the size of the subgraph traversed, not the overall graph size.

Another implication of graph databases is their flexibility. Behavioral scientists aim to connect data in ways that reflect their knowledge and expertise. This is why graph databases serve as the underlying infrastructure for constructing "knowledge graphs" (Barrasa & Webber, 2023). With these knowledge graphs, researchers allow the database to evolve alongside their understanding of the behavioral phenomenon rather than being rigidly defined upfront, when knowledge is most limited. This is particularly important when a behavioral phenomenon lacks theoretical background or lacks replication (Burgos, 2025). Graph databases fulfill this need by providing a flexible model that adapts to changing requirements and evidence. Because graphs are inherently additive, new nodes, relationships, labels, and subgraphs can be introduced. The modifications introduced to the original graph do not imply a threat to existing queries or application functionality. This flexibility minimizes the need for exhaustive upfront modeling and lowers the frequency of costly migrations (particularly for companies that hire behavior data scientists in charge of analyzing customers observed behavior), thereby reducing maintenance overhead in both academic and business settings.

A third implication of graph databases is the agility that they offer. Modern graph databases enable smooth development and easy system maintenance. Their schema-free design, combined with testable APIs and query languages, allows controlled evolution of applications. While the absence of rigid schemas means traditional governance mechanisms are missing, this is not a drawback. On the contrary, it encourages more transparent and actionable data governance. Typically, governance is enforced programmatically through tests that validate data models, queries, and business rules. This approach aligns well with agile and test-driven development practices, making graph database applications adaptable to changing business needs. In psychology, these aspects can be vital for the so-called "automated data collection techniques." For example, mobile applications, such as ecological momentary assessment tools, prompt users to report emotions or behaviors throughout the day, while motion sensors and radio frequency identification tags can monitor movement and location in homes or classrooms (Bak et al., 2021).

It is worth mentioning that relational databases acknowledge relationships, but only during modeling, where they serve as join mechanisms between tables. In graph databases, we often need to clarify the meaning of relationships and even qualify their strength. These are aspects that relational database management systems do not address explicitly (Robinson et al., 2015). From this viewpoint, graph databases demand both ontological and epistemological considerations like the ones recently described in the literature of theoretical and philosophical psychology (Burgos, 2025). Roughly speaking, ontological considerations relate to traditional questions of pure ontology like what is it to be? what does exist? What is the nature of what exist? In contrast, epistemological considerations pertains matters of evidence, explanation, certainty and truth (Burgos, 2025). As datasets grow more complex and less uniform (provided new evidence is available), relational data management systems like Microsoft Access, PostgreSQL, or SQLite become burdened with large join tables, sparsely populated rows, and extensive null-checking logic. Greater interconnectedness in relational databases leads to more joins, which degrade performance and complicate adaptation to evolving requirements. Soto (2025) has highlighted some of these limitations as "challenges to adoption." From the lenses of graph databases, however, these challenges are not even necessary. I will elaborate upon this particular aspect in the next section. As a wrap-up, labeled property graphs are not merely a technical alternative to relational schemas; they represent a methodological advance that aligns with psychology's need for flexible, scalable, and conceptually transparent data models.

4 Applying graph database in behavioral research

To illustrate the application of graph databases for behavioral scientists, I revisit the Speed-Dating dataset (Selterman et al., 2023) which provides a replication study from the seminal article by Eastwick and Finkel (2008). The general description of this replication study is available in an OSF repository. This repo provides access to the "speed.dating.2.zip'' file which contains the raw data in an SPSS data format (.sav). This file can be easily read in SPSS, PSPP, Jasp, Jamovi, or RStudio. Given the wide variety of graph database management systems in the market (e.g., Neo4j, Microsoft Azure Cosmos, Aerospike, Amazon Web Services Neptune, NebulaGraph, Memgraph, TigerGraph, Giraph), the rest of this work relies on Neo4j (Barrasa & Webber, 2023).

Neo4j ranks as a leading freemium software for graph database management systems and has been used in several industry sectors including retail, finance, pharmaceutics, hospitality, and electronic commerce, among others. Neo4j offers free and enterprise editions, which can be installed on local or server environments following official documentation. As the official documentation provides helpful material for newcomers, downloading and installation details are not necessary here, and the reader can consult specific details elsewhere (Van Bruggen, 2014).

4.1 Prerrequisites and materials

the Speed-Dating dataset (Selterman et al., 2023) which was a replication study from the original article by Eastwick and Finkel (2008).

4.2 STEP 1: Instance creation and data import

4.3 STEP 2: Understanding the context

4.4 STEP 3: Sketching a graph database model

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